

Wasseem Emad Anis Metry

ML Engineer

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Summary:

Machine Learning Engineer with hands-on experience in building real-world AI systems and LLM-powered applications. Skilled in Python, TensorFlow, and integrating ML models into web apps using FastAPI and Streamlit. Passionate about combining model accuracy with intuitive user experience.

SKILLS

- Languages: Python, SQL, MATLAB, MangoDB, C/C++.
- ML Frameworks: TensorFlow, PyTorch, scikit-learn.
- Tools: FastAPI, Streamlit, LangChain, Git, Mlflow (beginner), ZenML (beginner), LangGraph (beginner), crewAi (beginner).
- Concepts: Deep Learning, NLP.

Education:

Bachelor of Science MECHATRONICS ENGINEERING AIN SHAMS UNIVERSITY	2019 – 2024
DUAL DEGREE UEL B.SC. ENGINEERING HONS. University of East London	2020 – 2024

Experience:

3D Printing Market Lead Analyzer – Freelance Project

GPT-4 | Streamlit | Web Scraping | RAG | Prompt Engineering

Developed a full-stack AI tool that scrapes and analyzes recent 3D printing articles to identify companies likely to benefit from design automation software (e.g., Trinckle 3D).

- Built a web app using Streamlit with topic input and CSV download functionality.
- Used DuckDuckGo Search and BeautifulSoup to fetch and clean real-time article content.
- Designed structured prompts for GPT-4 to summarize articles, extract company insights, and rate lead potential (0–10).
- Implemented LLM logic to detect trends like parametric design, customization, and automation.
- Saved structured outputs into CSV for client use in business development.

Machine Learning Intern | Prodigy InfoTech | October 2024 – November 2024 ([certificate](#)):

- **Applied** linear regression to predict house prices based on key features, reducing RMSE from **48,768 to 0.29** after log transformation, significantly improving real estate valuation accuracy.
- Developed a K-means clustering algorithm from scratch to segment retail customers based on purchase history, enabling targeted marketing strategies.
- Built a support vector machine (SVM) to classify images of cats and dogs using the Kaggle dataset, achieving high classification accuracy, started with 68% accuracy then increase it to 78%.
- Implemented hand gesture recognition model using CNN with 99% validation accuracy.

Major Projects

- Built two versions of an end-to-end AI Resume-to-Job Matching System: a RESTful **FastAPI** version and an autonomous agent-based **CrewAI** version, using **spaCy**, **OpenAI GPT-4**, **MongoDB**, and **Pushover** for resume parsing, job scraping, and LLM-driven scoring.
- **Built a full-stack Amazon stock prediction app** using an LSTM model (TensorFlow) with a FastAPI backend, Streamlit frontend, and a GPT-3.5-powered LangChain agent for natural language Q&A on predicted prices, supporting CSV/NPY uploads, chart/table views, and user-driven analysis.

- **Built an LLM-powered assistant** using FastAPI that summarizes YouTube videos and answers user questions via transcript analysis and real-time web search, leveraging LangChain agents and the Mistral-7B model on Hugging Face.
- **Customer Churn Prediction System** – Developed an ML & LLM-powered chatbot for telecom customer retention, integrating a predictive model with a natural language interface to assess churn risk. Built the backend using **FastAPI** and the frontend using **Streamlit** for seamless interaction.
- **Language Detection & Translation App**: Developed a deep learning-based **language detection system** using an **LSTM model** capable of detecting **37 languages** with **92.2% test accuracy**. Built a **Streamlit web app** that detects languages and translates text using **Google Translator API**.
- **Image Captioning with Deep Learning (Flickr8k)** – Built an end-to-end image captioning model using **Xception CNN** for feature extraction and an **LSTM decoder** for sequence generation. Implemented top-k sampling for inference.
- **Developed a Machine Learning-Based Movie Recommender System** using **Collaborative Filtering** and **K-Nearest Neighbors (KNN)**, built an interactive **Streamlit web app**, and integrated **TMDb & OMDb APIs** to fetch movie posters. Optimized the **Movie Recommendation System dataset** using **Pandas & NumPy**, implemented **sparse matrix representation** for efficiency, and deployed the model using **Pickle serialization**.
- **Implemented a Retrieval-Augmented Generation (RAG) system** using LangChain and HuggingFace, and ChromaDB, enabling intelligent document retrieval from PDFs and context-aware conversational AI for answering user queries with enhanced accuracy.
- Implemented sentiment analysis models on the IMDB dataset using an Encoder Transformer made from scratch in PyTorch (88% accuracy) and an LSTM in TensorFlow (88.9%) for performance comparison.
- Developed a self-driving car system as a graduation project (GPA: 4.0), including lane detection using Canny edge detection and OpenCV (90% success rate), object detection with YOLOv8 (85% success rate), and obstacle distance estimation using the MiDaS depth estimation model.
- Developed MNIST digit classifiers using TensorFlow and scikit-learn (97.85% validation accuracy), as well as a custom implementation from scratch (88.5% accuracy) to compare performance and model design.

Major Specializations, Courses, & Diplomas

- Supervised Machine Learning: Regression and Classification, [Coursera](#), 2024.
- Advanced Learning Algorithms, [Coursera](#), 2024.
- The Ultimate MySQL Bootcamp: Go from SQL Beginner to Expert, [Udemy](#), 2024.
- Unsupervised Learning, Recommenders, Reinforcement Learning, [Coursera](#), 2024.
- Deep Learning A-Z 2024: Neural Networks, AI & ChatGPT Prize, [Udemy](#), 2024.
- Machine Learning A-Z: AI, Python & R + ChatGPT Prize [2024], [Udemy](#), 2024.
- Neural Networks and Deep Learning, [Coursera](#), 2024.
- Getting started with TensorFlow 2, [Coursera](#), 2024.
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization, [Coursera](#), 2024.
- Sequence Models, [Coursera](#), 2024.
- Generative AI with Large Language Models, [Coursera](#), 2024.
- Intro to NLP for AI, [365 DataScience](#), 2024.
- LLM Engineering: Master AI, Large Language Models & Agents, [Udemy](#), 2025

Language

- Arabic (Native speaker)
- English